

Practice Quiz: Body Science — Module 05: Action

Name: ____ Date: ____

Part A: Multiple Choice

1. In Active Inference, when your body takes an action (like shivering when cold), it is:

A) A random response B) Your body acting to make reality match its prediction of what conditions should be C) A sign that your body is broken D) Something that doesn't involve predictions

1. The "fight or flight" response — where your heart races, your muscles tense, and your breathing speeds up — is your body:

A) Panicking for no reason B) Taking coordinated action to prepare for a predicted threat, redirecting energy to the systems that need it most C) Shutting down all systems D) A response that only happens to animals, not humans

1. When you pull your hand away from a hot stove before you even feel the pain, this reflex is:

A) You making a conscious decision B) Your body's fastest action pathway — the spinal cord acts on a prediction of damage before the brain is fully involved C) An accident D) Something that doesn't count as an action

1. Physical exercise improves your brain function because:

A) Exercise has no effect on the brain B) Exercise increases blood flow, delivers more oxygen, and triggers the release of chemicals that help your brain build and update its models C) Your brain doesn't need oxygen D) Only mental exercise helps the brain

1. When you catch a ball, your brain predicts where the ball will be, and your hand moves there before the ball arrives. This shows that:

A) Your hand is smarter than your brain B) Your body's actions are driven by predictions about the future, not just reactions to the present C) You can only catch things by watching them the whole time D) Prediction has nothing to do with motor skills

1. When you learn to ride a bike, your actions start as wobbly and clumsy. Over time they become smooth. This happens because:

A) Your muscles grow bigger B) Your brain's prediction-action models become more accurate with practice, requiring less conscious effort C) The bike gets easier to ride D) Biking skill is something you're born with

1. Your pupils dilate (get bigger) in the dark. This is your body:

A) Breaking down B) Taking action to let in more light because it predicts it needs more visual information C) Doing something completely random D) An action that has no purpose

Part B: Short Answer

1. Compare the **reflex** of pulling your hand from a hot surface with the **deliberate action** of deciding to go for a run. How are both of these actions driven by predictions? What's different about the speed and awareness involved?
2. When athletes practice the same movement thousands of times, they're building what's sometimes called "muscle memory." Using Active Inference ideas, explain what's really happening in the brain. Why do practiced actions feel **effortless** compared to new ones?
3. Describe how your body's actions change when you get sick (sleeping more, eating less, getting a fever). How are these **actions** your body's way of trying to return to a predicted healthy state? What would happen if your body took no action at all?